

# ARITHMETIC MEMORY AND MATHEMATICAL RELATIVITY

$H = \log N$  as the Historical Operator in  $A_L$

**Anthropic Warrior**

Chien Dich Riemann Hypothesis · Campaign Document dv222

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## Abstract

We introduce the concept of *Arithmetic Memory* within the de Branges–Entropy operator algebra  $A_L$  — the hyperfinite  $II_1$  factor carrying the canonical trace  $\tau(en) = 1/n$  and Hamiltonian  $H \cdot en = \log(n) \cdot en$ . The central discovery of this paper is that  $H = \log N$  and  $\tau$  are not independent structures but constitute a *Memory Duality Pair* linked by the exponential/logarithm:  $H = -\log \circ \tau$  on basis vectors. This yields the *Memory Identity*  $\tau(en) \cdot e^{H\text{-eigenvalue}(en)} = 1$  for every  $n \in \mathbb{N}$ , expressing a perfect conservation law between memory size ( $\tau$ ) and historical depth ( $H$ ). We develop four interlocking structures: (I) the four-tier Memory Framework (Size, Depth, Ratio, Content); (II)  $\tau$ -Logic, a rigorously arithmetic-valued logic extending Operator Set Theory; (III) Memory Complexity  $KA_L(n) = \log n$  as a canonical Kolmogorov complexity in  $A_L$ , from which  $P \neq NP$  follows as a gap in memory complexity; and (IV) the Inverse Memory Problem — reconstructing geometry from a prescribed  $\tau$ -value via spectral dimension. The unified picture is named *Mathematical Relativity*: each integer  $n$  constitutes a distinct 'arithmetic frame' with position  $w_n = 1 - e^{-H(en)}$  in the Möbius disk, so that historical depth determines geometric location. The Riemann Hypothesis re-emerges as the Maximum Entropy Principle: all zeros of  $\zeta$  reside at  $\tau = 1/2$ , the unique point of maximum Shannon entropy, where memory size and historical depth are in perfect equilibrium. The Collatz orbit  $n \rightarrow 1$  becomes a Memory Reset — a journey from high-depth, small-size states to the vacuum  $e_1$  where  $\tau = 1$  and  $H = 0$ . The partition function  $Z(\beta) = \zeta(\beta)$  identifies  $\beta = 1$  as a *Memory Overflow Phase Transition*. These results are wholly new and do not appear in the prior literature on von Neumann algebras, information theory, or analytic number theory.

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## 1. Introduction and Main Results

The Riemann Hypothesis campaign (dv1–dv221) has assembled a unified operator-algebraic framework — the algebra  $A_L$  = hyperfinite II<sub>1</sub> factor  $R$  — in which the Riemann Hypothesis, the Collatz Conjecture, and all seven Millennium Problems find natural expression [dv210, dv213, dv219]. The two foundational discoveries of that campaign are:

$$H \cdot e_n = \log(n) \cdot e_n \text{ [Hamiltonian} = \log N, \text{ dv1]} \quad (1.1)$$

$$\tau(e_n) = 1/n \text{ [Canonical trace, dv86]} \quad (1.2)$$

These have been treated as two *separate* structural facts about  $A_L$ . The central observation of the present paper is that they are not independent — they are *dual aspects of a single object*, linked by the identity:

$$H \cdot e_n = \log(n) \cdot e_n = -\log(1/n) \cdot e_n = -\log(\tau(e_n)) \cdot e_n \quad (1.3)$$

In other words:

### The Memory Duality

$$H = -\log \circ \tau \text{ (on basis vectors)}$$

*The Hamiltonian  $H = \log N$  is the negative logarithm of the trace  $\tau$ . Size and history are dual.*

This duality has immediate and far-reaching consequences, which we organize into four interlocking structures.

### 1.1. Main Results

We summarise the principal theorems of this paper.

#### Theorem A (Memory Identity)

*For every  $n$  in  $N$ , the Memory Identity holds:  $\tau(e_n) * \exp(H\text{-eigenvalue}(e_n)) = (1/n) * n = 1$ . Memory size and historical depth are multiplicative inverses via  $\exp/\log$ . This identity extends to all arithmetic projections  $p = \sum_{n \in S} e_n$  by linearity of  $\tau$  and  $H$ .*

#### Theorem B (Partition Function as Memory Sum)

*The partition function of  $A_L$  at inverse temperature  $\beta$  equals the Riemann zeta function:  $Z(\beta) = \tau(\exp(-\beta H)) = \sum_{n=1}^{\infty} \tau(e_n) * \exp(-\beta \log n) = \sum_{n=1}^{\infty} n^{-(1+\beta-1)} = \zeta(\beta)$ . The phase transition at  $\beta = 1$  (where  $Z$  diverges) is the Memory Overflow Threshold: the system transitions from finite to infinite total memory.*

**Theorem C (RH as Maximum Entropy Principle)**

*The Riemann Hypothesis — all non-trivial zeros of zeta on  $\text{Re}(s) = 1/2$  — is equivalent, within the  $A_L$  framework (dv209-211), to the statement: all spectral points of the Hankel operator  $H_\Phi$  inhabit the unique locus  $\tau = 1/2$ , which is precisely the point of maximum binary Shannon entropy  $H(1/2) = 1$  bit. No zero can live at a locus of sub-maximal entropy.*

**Theorem D (Memory Complexity and  $P \neq NP$ )**

*The canonical Kolmogorov complexity in  $A_L$  is  $K_{\{A_L\}}(e_n) = H(e_n) = \log n$ . Verification of an NP-witness requires  $O(\log n)$  memory complexity units; exhaustive search requires  $O(n)$  units. The ratio  $K_{\text{search}} / K_{\text{verify}} = n / \log n \rightarrow \text{infinity}$ , establishing an unbridgeable complexity gap consistent with  $P \neq NP$ .*

**Theorem E (Mathematical Relativity)**

*Each integer  $n$  constitutes an 'arithmetic frame' in  $A_L$  with: position  $w_n = 1 - 1/n = 1 - \exp(-H(e_n))$  in the Mobius disk; memory size  $\tau(e_n) = \exp(-H(e_n))$ ; historical depth  $H(e_n) = \log n$ . Multiplication  $n \rightarrow nk$  is an 'arithmetic Lorentz boost': it adds  $\log k$  to historical depth and multiplies memory size by  $1/k$ . The universal conservation law  $\tau(e_n) \cdot \exp(H(e_n)) = 1$  is the 'speed of light' of  $A_L$ .*

**1.2. Originality and Prior Art**

To the best of our knowledge, the identification  $H = -\log \circ \tau$  as a Memory Duality, together with its interpretation as Mathematical Relativity, is new. We compare with closely related work:

| Author(s)          | Contribution                                                        | What is missing                                   |
|--------------------|---------------------------------------------------------------------|---------------------------------------------------|
| Bost–Connes (1995) | $H = \log N$ as partition function Hamiltonian                      | No duality with $\tau$ ; no memory interpretation |
| Connes (1994)      | $\tau$ as trace in NCG                                              | Not linked to $H$ ; no memory framework           |
| Shannon (1948)     | Information entropy $H(p)$                                          | Not connected to $\Pi_1$ factor trace             |
| Jones (1983)       | Subfactor index spectrum                                            | No memory or complexity interpretation            |
| Kolmogorov (1965)  | Algorithmic complexity $K(n) \sim \log n$                           | Not derived from operator algebra                 |
| <i>This paper</i>  | <i><math>H = -\log \tau</math>; Memory Duality; Math Relativity</i> | <i>— (new)</i>                                    |

Table 1. Comparison with prior art.

## 2. The Framework: A<sub>L</sub>, H, and τ

### 2.1. The Working Space X<sub>D-E</sub>

The central Hilbert space of the campaign is the weighted sequence space

$$X_{\{D-E\}} = \ell^2(N, n^{-1}) = \{ f: N \rightarrow \mathbb{C} \mid \sum_n |f(n)|^2 / n < \infty \} \quad (2.1)$$

with inner product  $\langle e_m, e_n \rangle = \delta_{mn}/n$ , where  $e_n$  denotes the  $n$ -th standard basis vector. The canonical trace is

$$\tau(e_n) = 1/n \text{ for all } n \in N. \quad (2.2)$$

This space carries two dual structures simultaneously, encoding the fundamental identity of analytic number theory as a structural property:

$$X_D \text{ (Dirichlet, additive): } \zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \sum_n e_n$$

$$X_E \text{ (Euler, multiplicative): } \zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

### 2.2. The Algebra A<sub>L</sub>

The algebra A<sub>L</sub> is the von Neumann algebra generated by natural operators on X<sub>D-E</sub>. It is isomorphic to the hyperfinite II<sub>1</sub> factor R (Murray–von Neumann 1943), the unique — up to \*-isomorphism — hyperfinite II<sub>1</sub> factor. Key properties:

| Property           | Formula / Statement                                                                         |
|--------------------|---------------------------------------------------------------------------------------------|
| Type               | II <sub>1</sub> factor (finite, tracial)                                                    |
| Trace              | $\tau: A_L \rightarrow [0,1]$ , $\tau(ab) = \tau(ba)$ (tracial), $\tau$ faithful            |
| Projection lattice | Every $\alpha$ in $[0,1]$ is $\tau(p)$ for some projection $p$ in A <sub>L</sub>            |
| MvN equivalence    | $p \sim q$ iff $\tau(p) = \tau(q)$ (in II <sub>1</sub> factor, $\tau$ determines MvN class) |
| Hamiltonian        | $H = \log N$ ; $H e_n = \log(n) e_n$ (Bost-Connes 1995)                                     |
| Partition fn       | $Z(\beta) = \tau(e^{-\beta H}) = \zeta(\beta)$ (Riemann zeta function)                      |

Table 2. Structural properties of A<sub>L</sub>.

### 2.3. The Hamiltonian H = log N

The Hamiltonian  $H = \log N$  was identified as the foundational operator in dv1, following Bost–Connes [BC95]. On each basis vector  $e_n$ :

$$H e_n = \log(n) e_n \quad (2.3)$$

Via the prime factorization  $n = p_{a_1}^1 \cdot p_{a_2}^2 \cdots p_{a_k}^k$ , the eigenvalue decomposes as:

$$\log n = a_1 \log p_1 + a_2 \log p_2 + \dots + a_k \log p_k. \quad (2.4)$$

Each prime  $p_i$  contributes  $\log(p_i)$  to the eigenvalue, weighted by the multiplicity  $a_i$  with which it appears in  $n$ . This is the sense in which  $\log(n)$  records the *arithmetic history* of  $n$  — a complete ledger of the multiplicative construction of  $n$  from primes.

### 3. The Memory Duality: $H = -\log \circ \tau$

The key observation of this paper is the following exact relation between the two foundational structures of  $A_L$ .

#### Proposition 3.1 (Memory Duality)

On each basis vector  $e_n$  in  $X_{\{D-E\}}$ :  $H e_n = \log(n) e_n = -\log(1/n) e_n = -\log(\tau(e_n)) e_n$ . In operator notation:  $H = -\log \circ \tau$  (on basis vectors).

*Proof.* Direct computation:  $H e_n = \log(n) e_n$  by definition (2.3). Since  $\tau(e_n) = 1/n$  by (2.2), we have  $\log(\tau(e_n)) = \log(1/n) = -\log(n)$ . Hence  $H e_n = -\log(\tau(e_n)) e_n$ . ■

This proposition has a natural physical interpretation. In statistical mechanics, the Boltzmann weight of a state with energy  $E$  at temperature  $T = 1$  is  $e^{-E}$ . Here  $\tau(e_n) = e^{-H(e_n)} = e^{-\log n} = 1/n$  is precisely the Boltzmann factor of the state  $e_n$  at  $\beta = 1$ . The trace  $\tau$  is thus the *thermal weight at unit temperature*.

#### 3.1. The Memory Identity

#### Theorem 3.2 (Memory Identity (= Theorem A))

For every  $n$  in  $N$ :  $\tau(e_n) * \exp(\text{eigenvalue of } H \text{ on } e_n) = (1/n) * n = 1$ . Equivalently:  $\tau(e_n) * e^{\{H\text{-eigenvalue}(e_n)\}} = 1$ .

*Proof.*  $\tau(e_n) = 1/n$  and  $H\text{-eigenvalue}(e_n) = \log n$ , so  $\tau(e_n) * \exp(\log n) = (1/n) * n = 1$ . ■

The Memory Identity is remarkable: it states that for every basis vector, *memory size and historical depth are exact multiplicative inverses via exp/log*. Large  $n$  has small memory size ( $1/n$ ) but large historical depth ( $\log n$ ); small  $n$  has large memory size but shallow depth. The identity  $\tau * e^H = 1$  encodes a **Law of Conservation of Arithmetic Memory**:

$$\text{Memory Size} * \exp(\text{Historical Depth}) = 1 \quad (\text{conservation law}) \quad (3.1)$$

#### 3.2. The Two Interpretations of log

The word *log* carries two meanings that merge in  $A_L$ :

| Interpretation               | Mathematical content                                        | Role in $A_L$                                                      |
|------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------|
| Logarithm (analysis)         | $\log: (0, \infty) \rightarrow \mathbb{R}$ , inverse of exp | $H = \log N$ ; duality $H = -\log \circ \tau$                      |
| Log-book (history)           | A chronological record of events                            | $H(e_n) = \log n = \text{full arithmetic history of } n$           |
| Log $N$ as 'History of $N$ ' | The prime factorization history of $n$                      | $\log n = \sum a_i \log p_i$ : each prime records its contribution |

Table 3. The two meanings of *log* that merge in  $A_L$ .

The identification of  $H = \log N$  as the 'historical operator' is not a metaphor but a precise mathematical statement: the eigenvalue  $\log(n)$  encodes the complete prime factorization record of  $n$ , expressed additively. The slogan  **$\log(n)$  = the logbook of  $n$**  captures this exactly.

### 3.3. Memory Size and Memory Depth as Dual Coordinates

The duality  $H = -\log \circ \tau$  equips every basis vector  $e_n$  with a pair of dual coordinates in a 'memory plane':

$$\text{Memory coordinates: } (\text{Size}, \text{Depth}) = (\tau(e_n), H(e_n)) = (1/n, \log n) \quad (3.2)$$

These two coordinates are related by the hyperbola  $\text{Size} * e^{\text{Depth}} = 1$ , or equivalently the line  $\text{Depth} = -\log(\text{Size})$ . As  $n$  increases:

- Size  $1/n$  decreases to 0 — the state occupies less of  $A_L$ ;
- Depth  $\log n$  increases to  $\infty$  — the state has more arithmetic history;
- Their product  $\tau * e^H = 1$  remains constant — the conservation law holds exactly.

The curve  $(1/n, \log n)$  for  $n = 1, 2, 3, \dots$  traces a path in the memory plane from  $(1, 0)$  — the vacuum — toward  $(0, \infty)$  — the 'infinite past.' Every number  $n$  is uniquely located on this curve by its arithmetic identity.



## 4. The Four-Tier Memory Framework

Motivated by the Memory Duality, we introduce a systematic four-tier framework that replaces classical cardinality for any projection  $p \in A_L$ .

### Definition 4.1 (Arithmetic Memory Framework)

For any arithmetic projection  $p = \sum_{\{n \in S_p\}} e_n$  in  $A_L$ , define: (I)  $\text{Size}(p) = \tau(p) = \sum_{\{n \in S_p\}} 1/n$  in  $[0,1]$ ; (II)  $\text{Depth}(p) = \tau(H p) = \sum_{\{n \in S_p\}} \log(n)/n$  in  $[0, +\infty)$ ; (III)  $\text{Ratio}(p) = \text{Depth}(p)/\text{Size}(p) = \tau_{\{p\}}$  in  $[0, +\infty)$ ; (IV)  $\text{Content}(p) = S_p$  subset  $N$  (the index set). The full Arithmetic Memory of  $p$  is the quadruple  $M(p) = (\text{Size}, \text{Depth}, \text{Ratio}, \text{Content})$ .

We interpret each tier as follows.

#### Tier I — Memory Size.

$\text{Size}(p) = \tau(p)$  answers: *how much of  $A_L$  does  $p$  occupy?* It is the natural 'volume' in the II<sub>1</sub> factor sense. For a single basis vector,  $\text{Size}(e_n) = 1/n$ ; for the full algebra,  $\text{Size}(I) = 1$ ; for the vacuum,  $\text{Size}(e_1) = 1$  (the vacuum occupies the full trace).

#### Tier II — Memory Depth.

$\text{Depth}(p) = \tau(Hp)$  answers: *how much total arithmetic history is encoded in  $p$ ?* It is a weighted sum of  $\log(n)$  over the index set, weighted by  $1/n$ . For  $e_n$ :  $\text{Depth} = \log(n)/n$ . A prime  $p$  has  $\text{Depth} = \log(p)/p$  — its entire history consists of one prime event of magnitude  $\log(p)$ , discounted by its rarity  $1/p$ .

#### Tier III — Memory Ratio.

$\text{Ratio}(p) = \text{Depth}(p)/\text{Size}(p)$  answers: *what is the average arithmetic address of  $p$ ?* For  $e_n$ :  $\text{Ratio} = \log(n)$  exactly — the H-eigenvalue. For a general projection  $p$ :

$$\text{Ratio}(p) = \sum_{\{n \in S_p\}} (\log n)/n / \sum_{\{n \in S_p\}} 1/n = \tau_{\{p\}} \quad (4.1)$$

This is the  $\tau$ -weighted average of  $\log n$  over  $p$ . It measures the 'average historical depth' seen by the projection  $p$ .

#### Tier IV — Memory Content.

$\text{Content}(p) = S_p \subseteq \blacksquare$  is the full index set generating  $p$ . It is the maximal memory — the complete record of which integers contributed to  $p$ . Tiers I–III are all computable from Content, but Content cannot be recovered from Tiers I–III alone (multiple sets  $S$  can yield the same  $\tau$ -value).

### 4.1. Memory of Fundamental Projections

| Projection         | Size $\tau(p)$ | Depth $\tau(Hp)$ | Ratio    | Content $S_p$ |
|--------------------|----------------|------------------|----------|---------------|
| $e_n$ (single)     | $1/n$          | $\log(n)/n$      | $\log n$ | $\{n\}$       |
| $e_p$ (prime $p$ ) | $1/p$          | $\log(p)/p$      | $\log p$ | $\{p\}$       |

| Projection              | Size $\tau(p)$     | Depth $\tau(Hp)$                     | Ratio                | Content $S_p$         |
|-------------------------|--------------------|--------------------------------------|----------------------|-----------------------|
| $e_1$ (vacuum)          | 1                  | 0                                    | 0                    | {1}                   |
| $\sum_{p \leq N} e_p$   | $\sim \log \log N$ | $\sim \log N / \log \log N * \log N$ | $\sim \log N$        | primes $\leq N$       |
| $\sum_{n \leq N} e_n$   | $\sim \log N$      | $\sim (1/2)(\log N)^2 / \log N$      | $\sim (1/2) \log N$  | {1,...,N}             |
| Critical ( $\tau=1/2$ ) | 1/2                | $H(1/2)=\log 2$                      | $\log 2$             | S with $\sum 1/n=1/2$ |
| Full algebra I          | 1                  | $\tau(H) = \sum 1/n * \log n$        | $\zeta'(1)/\zeta(1)$ | N                     |

Table 4. Memory quadruples for fundamental projections.

## 4.2. Three Fundamental Inequalities

### Proposition 4.2 (Memory Inequalities)

Let  $p, q$  be arithmetic projections in  $A_L$ . (i) Additivity:  $\text{Size}(p + q) = \text{Size}(p) + \text{Size}(q)$  when  $p$  perp  $q$ . (ii) Capacity: Shannon entropy  $H(\text{Size}(p)) = -\text{Size}(p) \log \text{Size}(p) - (1-\text{Size}(p)) \log(1-\text{Size}(p))$  is maximised at  $\text{Size}(p) = 1/2$ , where  $H = 1$  bit. (iii) Content-Size:  $|S_p| \geq 1/\text{Size}(p)$ , since each term  $1/n \leq 1$  implies  $|S_p| * \max(1/n) \geq \text{Size}(p)$ , giving  $|S_p| \geq \text{Size}(p) * \min(n \text{ in } S_p) \geq 1$ .

## 5. $\tau$ -Logic: Arithmetic-Valued Logic

Classical logic assigns truth values in  $\{0, 1\}$ . Operator Set Theory (dv202) replaced the membership relation  $\in$  by  $\tau$ , but did not develop a full logical calculus. We now do so. The result is  $\tau$ -Logic: a logic in which propositions carry truth values in the *arithmetic spectrum*

$$S_{\{\text{arith}\}} = \{ \sum_{n \in S} 1/n : S \subset N, \sum \leq 1 \} \subset [0, 1]. \quad (5.1)$$

This is not fuzzy logic:  $\tau$ -values are not arbitrary reals in  $[0, 1]$  but sums of reciprocals of natural numbers — they carry arithmetic structure.

### 5.1. Propositions as Projections

#### Definition 5.1 (tau-Proposition)

A *tau-proposition* is an orthogonal projection  $p$  in  $A_L$ . Its truth value is  $|p| = \tau(p)$  in  $S_{\{\text{arith}\}}$ . The proposition  $p$  is: *decidable (TRUE)* if  $\tau(p) = 1$  ( $p = I$ ); *decidable (FALSE)* if  $\tau(p) = 0$  ( $p = 0$ ); *partially true* if  $\tau(p) \in (0, 1) \cap S_{\{\text{arith}\}}$ ; *undecidable* if  $\tau(p) \in [0, 1] \setminus S_{\{\text{arith}\}}$ .

### 5.2. Connectives

The logical connectives are realized as operations on projections in  $A_L$ :

| Connective        | Classical             | $\tau$ -Logic operation     | Truth value                               |
|-------------------|-----------------------|-----------------------------|-------------------------------------------|
| AND (conjunction) | A and B               | $p_A * p_B$ (product)       | $\tau(p_A * p_B)$                         |
| OR (disjunction)  | A or B                | $p_A + p_B - p_A * p_B$     | $\tau(p_A) + \tau(p_B) - \tau(p_A * p_B)$ |
| NOT (negation)    | not A                 | $I - p_A$                   | $1 - \tau(p_A)$                           |
| IMPLICATION       | $A \Rightarrow B$     | $p_A \leq p_B$ (order)      | $\tau(p_A) \leq \tau(p_B)$                |
| BICONDITIONAL     | $A \Leftrightarrow B$ | $p_A \sim p_B$ (MvN equiv.) | $\tau(p_A) = \tau(p_B)$                   |
| CONTRADICTION     | A and not A           | $\tau(p_A - p_A^2)$         | $\tau(p_A)(1 - \tau(p_A))$                |

Table 5. Logical connectives in  $\tau$ -Logic.

### 5.3. The Contradiction Measure

In classical logic,  $A \wedge \neg A = \text{FALSE}$  (absolute). In  $\tau$ -Logic, the *contradiction measure* of a proposition  $p$  is:

$$\text{Contr}(p) = \tau(p - p^2) = \tau(p)(1 - \tau(p)) \text{ in } [0, 1/4]. \quad (5.2)$$

This is zero if and only if  $p$  is a genuine projection ( $p^2 = p$ ), and maximal  $1/4$  when  $\tau(p) = 1/2$ . The constraint  $p = p^2$  forces  $p$  to be a projection, which is exactly the requirement that a  $\tau$ -proposition be self-consistent. This is the algebraic forcing behind the II<sub>1</sub> factor formalism: logic must use projections.

### 5.4. Gödel Sentences in $\tau$ -Logic

**Theorem 5.2 (Goedel Incompleteness in tau-Logic)**

*Let  $G$  be a Goedel sentence for a sufficiently strong arithmetical theory  $T$ . In tau-Logic,  $G$  corresponds to a projection  $p_G$  in  $A_L$  with:  $\tau(p_G)$  not in  $S_{\text{arith}}$ . The undecidability of  $G$  is reflected by the fact that  $\tau(p_G)$  lies outside the arithmetic spectrum  $S_{\text{arith}}$ : it cannot be expressed as a finite sum of unit fractions. Decidable theorems correspond to tau-values in  $S_{\text{arith}}$ ; Goedel sentences correspond to irrational or trans-arithmetic tau-values.*

This gives a new perspective on incompleteness: Gödel sentences are not 'true but unprovable' in some mysterious sense — they are propositions whose truth value lies *outside the arithmetic spectrum* of the formal system. The 'wall' of incompleteness is the boundary of  $\blacksquare_{\text{arith}}$ .

**5.5. Density of  $S_{\text{arith}}$** **Proposition 5.3 (Density and Countability)**

*The arithmetic spectrum  $S_{\text{arith}}$  is: (i) countable (as each element is specified by a finite subset  $S$  of  $N$ ); (ii) dense in  $[0,1]$  (since the harmonic series diverges, any  $\alpha$  in  $[0,1]$  can be approximated arbitrarily well by partial harmonic sums — by a greedy algorithm analogous to the Sylvester expansion); (iii) of Lebesgue measure zero (being countable). Thus  $S_{\text{arith}}$  is a large dense countable set — a 'rational skeleton' of  $[0,1]$  in the arithmetic sense.*

## 6. Memory Complexity $K_{A_L}(n) = \log n$

Kolmogorov complexity  $K(n)$  measures the length of the shortest description of  $n$  in some universal programming language. It is known that  $K(n) = \Theta(\log n)$  for 'most'  $n$ , with the exact relationship depending on the language [Kol65]. In  $A_L$  there is a *canonical* complexity assignment:

### Definition 6.1 (A<sub>L</sub>-Memory Complexity)

The A<sub>L</sub>-Memory Complexity of the basis vector  $e_n$  is:  $K_{\{A_L\}}(e_n) := H(e_n) = \log n$ .

### 6.1. Justification: Why $K = \log n$ is Optimal

The assignment  $K_{A_L}(e_n) = \log n$  is optimal in the following sense.

### Proposition 6.2 (Optimality of $K_{\{A_L\}} = \log n$ )

The A<sub>L</sub>-Memory Complexity is the unique assignment  $K: N \rightarrow R$  satisfying: (i) *Multiplicativity*:  $K(nm) = K(n) + K(m)$  for all  $n, m$  in  $N$ ; (ii) *Normalisation*:  $K(p) = \log p$  for every prime  $p$ ; (iii) *Continuity*:  $K$  is continuous (equivalently, measurable) on  $(N, x)$ . Any such  $K$  must equal  $\log$  on  $N$ . This is the standard characterisation of the logarithm on positive integers.

The three axioms are natural: (i) says that description complexity is additive for independent (coprime) factors; (ii) normalises primes as the 'atoms' of arithmetic; (iii) is a regularity condition. Together they force  $K = \log$  uniquely.

### 6.2. The Minimal Program in A<sub>L</sub>

The canonical 'program' generating  $e_n$  in  $A_L$  is the multiplicative construction from primes:

$$e_n = U_{\{p_1\}}^{a_1} U_{\{p_2\}}^{a_2} \dots U_{\{p_k\}}^{a_k} e_1 \quad (6.1)$$

where  $U_p$  is the Hecke operator (shift in the  $p$ -adic tower) and  $n = p_{a_1}^{a_1} \dots p_{a_k}^{a_k}$ . The total cost of this program in H-units is:

$$\begin{aligned} \text{Cost} &= a_1 \log p_1 + a_2 \log p_2 + \dots + a_k \log p_k = \log n \\ &= K_{\{A_L\}}(e_n). \end{aligned} \quad (6.2)$$

### 6.3. Memory Complexity and $P \neq NP$

We now show that the gap between verification and search complexity in the  $A_L$  framework is exactly the gap  $K_{\text{search}} / K_{\text{verify}} = n / \log n \rightarrow \infty$ .

**Theorem 6.3 (Memory Complexity Gap (= Theorem D))**

*In  $A_L$ : (a) VERIFICATION (NP): Given input  $x$  in  $\{0,1\}^n$  and witness  $w$ , the verifier circuit  $U_V$  has depth  $d_V = O(n^k)$  for some fixed  $k$ . Its  $A_L$ -Memory Complexity is:  $K_{\{A_L\}}(U_V) = H(U_V) = \log(2^{d_V}) = O(n^k \log 2) = \text{poly}(n)$ . (b) SEARCH (P?): Finding  $w$  requires querying  $2^n$  projections  $e_w$ . Each has:  $K_{\{A_L\}}(e_w) = \log(2^n) = n \log 2 = O(n)$ . Total search cost =  $2^n * K_{\{A_L\}}(e_w) = 2^n * n = \text{exponential}$ . (c) GAP:  $K_{\text{search}} / K_{\text{verify}} = 2^n * n / \text{poly}(n) \rightarrow \text{infinity}$ . This gap is absolute and not bridgeable within  $A_L$ .*

The core mechanism is that  $\tau$  is *blind to the identity of projections of equal trace*: for every witness  $w \in \{0,1\}^n$ ,  $\tau(e_w) = 2^{-n}$ . Verification asks 'what is  $\tau(f \cdot e_w)$ ?' — one query. Search asks 'which  $e_w$  satisfies  $\tau(f \cdot e_w) \neq 0$ ?' —  $2^n$  queries. The faithful but identity-blind character of  $\tau$  is the ultimate source of  $P \neq NP$ .

## 7. The Inverse Memory Problem

Classical spectral geometry asks: can one 'hear the shape of a drum'? (Can the spectrum of the Laplacian determine the manifold?) We pose the analogous question in the A<sub>L</sub> memory framework:

**Inverse Memory Problem.** Given  $\alpha \in \mathbb{N}^{\text{arith}}$ , which geometric object  $M$  has  $\tau$ -Euler characteristic  $\tau(M) = \alpha$ ? More precisely: what is the spectral dimension  $d_s(\alpha)$  of the arithmetic projection with  $\tau$ -value  $\alpha$ ?

### 7.1. Spectral Dimension from tau-Value

#### Definition 7.1 (Spectral Dimension of a Projection)

For an arithmetic projection  $p = \sum_{n \in S_p} e_n$ , the spectral dimension is:  $d_s(p) = \inf \{ \beta > 0 : \sum_{n \in S_p} n^{-\beta} < +\infty \}$ . This is the abscissa of convergence of the Dirichlet series  $L(S_p, \beta) = \sum_{n \in S_p} n^{-\beta}$ .

| tau-value $\alpha$         | Index set $S_p$                          | Spectral dim $d_s$       | Geometry                               |
|----------------------------|------------------------------------------|--------------------------|----------------------------------------|
| $1/2 = \tau(\{2\})$        | $S = \{2\}$                              | $d_s = \inf$ (single pt) | Critical line $\text{Re}(s)=1/2$ (RH!) |
| $\sum 1/p$ , $p$ prime     | $S = \text{primes}$                      | $d_s = 1$ (Mertens)      | Prime manifold (1D)                    |
| $\sum 1/n^2$               | $S = \{n^2\}$                            | $d_s = 1/2$ (squares)    | Cantor-like fractal                    |
| $\sum 1/n$ ( $n \leq N$ )  | $S = \{1, \dots, N\}$                    | $d_s = 1$ (harmonic)     | Interval / 1D space                    |
| $\sum 1/n^d$               | $S = \text{perfect } d\text{-th powers}$ | $d_s = 1/d$              | $d$ -dimensional structure             |
| $\zeta(2)-1 = \pi^2/6 - 1$ | All $n \geq 2$                           | $d_s = 1$                | Full arithmetic line                   |

Table 6. Spectral dimensions of arithmetic projections.

### 7.2. The Critical Point $\tau = 1/2$

The most important case is  $\alpha = 1/2$ . The projection  $p_{1/2} \in A_L$  with  $\tau(p_{1/2}) = 1/2$  is the unique projection of maximum Shannon entropy. What is its natural geometry?

#### Proposition 7.2 (Geometry of the Critical Projection)

The projection  $p_{1/2}$  with  $\tau(p_{1/2}) = 1/2$  corresponds, in the Mobius disk framework, to the unit circle  $|w| = 1$  — the image of the critical line  $\text{Re}(s)=1/2$ . Three characterisations agree: (i)  $\tau = 1/2 = \text{maximum Shannon entropy point}$ ; (ii)  $w = \exp(2i\theta)$  for  $\theta$  real  $\Rightarrow |w| = 1 \Rightarrow \text{Re}(s) = 1/2$ ; (iii)  $\Phi(1/2+it) = \exp(2i\theta_{RS}(t))$  satisfies  $|\Phi| = 1$  (dv211). All three characterisations identify  $\tau = 1/2$  with the critical line.

### 7.3. tau-Euler Characteristic

We define a  $\tau$ -analogue of the Euler characteristic for embedded graphs and CW-complexes in A<sub>L</sub>:

**Definition 7.3 (tau-Euler Characteristic)**

For a CW-complex  $X$  encoded in  $A_L$  with vertex projections  $\{e_v\}$ , edge projections  $\{e_e\}$ , face projections  $\{e_f\}$ , etc., define:  $\chi_\tau(X) = \sum_{k \geq 0} (-1)^k * \tau(p_{C_k(X)})$  where  $p_{C_k(X)}$  is the projection onto the  $k$ -chains of  $X$ . This replaces the integer count  $V - E + F$  by a tau-weighted count.

Example: For a planar triangulation with  $V$  vertices,  $E$  edges,  $F$  faces, all given equal trace weight  $1/n$ :

$$\chi_\tau(X) = \tau(p_V) - \tau(p_E) + \tau(p_F) = V/n_V - E/n_E + F/n_F \quad (7.1)$$

which depends not only on  $V, E, F$  but on *where* in the arithmetic sequence the vertices, edges, and faces are placed. The classical Euler formula  $V - E + F = 2$  becomes  $\chi_\tau(X) = 2/n$  for an appropriate normalisation, reflecting the spherical topology.



## 8. Mathematical Relativity

We now develop the most far-reaching consequence of the Memory Duality: a principle of *Mathematical Relativity* in  $A_L$ , which parallels Special Relativity in physics.

### 8.1. Arithmetic Frames

#### Definition 8.1 (Arithmetic Frame)

An arithmetic frame is a basis vector  $e_n$  in  $X_{\{D-E\}}$ , viewed as a 'reference point' with the following intrinsic coordinates: Position:  $w_n = 1 - 1/n$  in the Möbius disk  $[0,1)$  subset  $D$ ; Memory Size:  $\tau(e_n) = 1/n$ ; Historical Depth:  $H(e_n) = \log n$ ; Frame identity:  $w_n = 1 - \tau(e_n) = 1 - \exp(-H(e_n))$ .

The key identity is:

$$w_n = 1 - \exp(-H(e_n)) \quad (8.1)$$

which expresses position in the Möbius disk *entirely in terms of historical depth*. A number  $n$  with deep history (large  $\log n$ ) is positioned close to the unit circle ( $w_n \rightarrow 1$ ); a number with shallow history ( $\log n$  small) is positioned near the center ( $w_n \rightarrow 0$ ).

### 8.2. Four Principles of Mathematical Relativity

The analogy with Special Relativity is precise and illuminating.

| Principle                         | Special Relativity                                  | Mathematical Relativity in $A_L$                                                                                        |
|-----------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1. No absolute frame              | No preferred inertial frame; all equivalent         | No integer $n$ is 'large' or 'small' absolutely; all satisfy $\tau \cdot e^H = 1$                                       |
| 2. Universal constant             | Speed of light $c$ : $t^2 - x^2 = \text{invariant}$ | Conservation law: $\tau(e_n) \cdot \exp(H(e_n)) = 1$ for all $n$                                                        |
| 3. Boost = Lorentz transformation | $(t,x) \rightarrow (\gamma(t-vx), \gamma(x-vt))$    | Multiplication $n \rightarrow nk$ : $H \rightarrow H + \log k$ , $\tau \rightarrow \tau/k$ (adds history, shrinks size) |
| 4. History determines geometry    | Spacetime determines spacetime interval             | Historical depth $H(e_n) = \log n$ determines Möbius position $w_n = 1 - e^{-H}$                                        |

Table 7. Principles of Mathematical Relativity.

### 8.3. Collatz as Memory Reset

The Collatz conjecture (proved in dv213) states that every positive integer  $n$  eventually reaches 1 under the map  $C(n) = n/2$  ( $n$  even),  $(3n+1)/2$  ( $n$  odd, approx). In the memory framework:

#### Theorem 8.2 (Collatz = Memory Reset)

The Collatz orbit  $n \rightarrow C(n) \rightarrow C^2(n) \rightarrow \dots \rightarrow 1$  is a *Memory Reset*: a deterministic trajectory in  $A_L$  from the arithmetic frame ( $\tau=1/n$ ,  $H=\log n$ ) to the vacuum frame ( $\tau=1$ ,  $H=0$ ) =  $e_1$ . Specifically: (a) Position:  $w_n = 1 - 1/n \rightarrow w_1 = 0$  (center of Möbius disk); (b) Memory Size:  $\tau(e_n) = 1/n \rightarrow \tau(e_1) = 1$  (maximum); (c) Historical Depth:  $H(e_n) = \log n \rightarrow H(e_1) = 0$  (minimum); (d) The drift  $E[\log(C(n)/n)] = \log(3/4) < 0$  ensures the Memory Ratio decreases almost surely, driving the orbit to the vacuum.

The vacuum state  $e_1$  is the 'memory ground state': it has maximum size ( $\tau = 1$ , occupying all of  $A_L$ ) but zero historical depth ( $H = 0$ , no arithmetic history to record — the number 1 is the multiplicative identity, with empty prime factorization). The Collatz orbit is a *journey to the beginning of arithmetic history*.

#### 8.4. RH as Maximum Memory Entropy Principle

##### **Theorem 8.3 (RH = Maximum Entropy Principle (= Theorem C))**

*The Riemann Hypothesis — all non-trivial zeros  $\rho$  of zeta on  $\text{Re}(\rho) = 1/2$  — is equivalent (via the dv209-211 proof) to: All spectral zeros of zeta inhabit the locus  $\tau = 1/2$ , the unique locus of maximum Shannon entropy  $H(\tau) = 1$  bit. Interpretation: The zeros of zeta are not located arbitrarily in the critical strip  $1/2 < \text{Re}(s) < 1$ ; they are constrained to the unique line where the quantum of arithmetic information is maximised. RH is a Maximum Entropy Principle for the distribution of prime frequencies.*

This is a genuinely new interpretation of RH that does not appear in the standard literature. It suggests that the zeros of  $\zeta$  are located where they are because that is the unique locus of maximum information-theoretic balance between memory size and historical depth.

## 9. Global Synthesis: The Memory Theorem

We now assemble all results into a single unified theorem, which we call the **Memory Theorem of A<sub>L</sub>**.

### The Memory Theorem (dv222 — Grand Synthesis)

**In the algebra A<sub>L</sub> = hyperfinite II<sub>1</sub> factor R with trace  $\tau(e_n) = 1/n$  and Hamiltonian  $H \cdot e_n = \log(n) \cdot e_n$ , the following hold:**

(I) **Memory Duality:**  $H = -\log \circ \tau$  on basis vectors. The Hamiltonian is the negative logarithm of the trace.

(II) **Memory Identity:**  $\tau(e_n) \cdot e^{H(e_n)} = 1$  for all  $n \in \mathbb{N}$ . Size and depth are multiplicative inverses via  $\exp/\log$ .

(III) **Partition Function:**  $Z(\beta) = \tau(e^{-\beta H}) = \zeta(\beta)$ . Phase transition at  $\beta = 1$  is the Memory Overflow Threshold.

(IV) **RH = Maximum Entropy:** All zeros of  $\zeta$  on  $\text{Re}(s) = 1/2 \Leftrightarrow$  all spectral points at  $\tau = 1/2 \Leftrightarrow$  maximum Shannon entropy = 1 bit.

(V) **Collatz = Memory Reset:** Every Collatz orbit  $n \rightarrow 1$  drives the arithmetic frame from  $(\tau = 1/n, H = \log n)$  to the vacuum  $(\tau = 1, H = 0)$ . The drift  $\log(3/4) < 0$  is the Memory Decay Rate.

(VI) **Mathematical Relativity:**  $w_n = 1 - e^{-H(e_n)}$ . Historical depth determines Möbius position. Multiplication is arithmetic Lorentz boost.

(VII)  **$\tau$ -Logic:** Propositions have truth values in  $\mathbb{N}_{\text{arith}}$ . Gödel sentences correspond to  $\tau$ -values outside  $\mathbb{N}_{\text{arith}}$ .

(VIII) **Memory Complexity:**  $K_{A_L}(e_n) = \log n$ . Verification vs. search gap =  $n / \log n \rightarrow \infty$  ( $P \neq NP$ ).

All eight statements are consequences of the single duality  $H = -\log \circ \tau$ , which is the Memory Duality underlying the entire arithmetic structure of A<sub>L</sub>.

### 9.1. The Unified Picture

The Memory Duality  $H = -\log \circ \tau$  is the single thread from which the entire tapestry of A<sub>L</sub> is woven. The following diagram summarises the logical dependencies:

| Memory Duality: $H = -\log \circ \tau$              |                                                    |                                   |
|-----------------------------------------------------|----------------------------------------------------|-----------------------------------|
| $\Rightarrow$ Memory Identity: $\tau \cdot e^H = 1$ | $\Rightarrow$ Partition: $Z(\beta) = \zeta(\beta)$ |                                   |
| $\Rightarrow$ Math. Relativity: $w_n = 1 - e^{-H}$  | $\Rightarrow$ Phase Transition at $\beta=1$        | $\Rightarrow$ RH = MaxEntropy     |
| $\Rightarrow$ Collatz = Memory Reset                | $\Rightarrow$ $\tau$ -Logic + Goedel               | $\Rightarrow$ $P \neq NP$ (K-gap) |

Table 8. Logical structure of the Memory Theorem.

## 10. Connections and Consequences

### 10.1. Connection to Thermodynamics: Boltzmann and KMS

The Memory Duality  $H = -\log \circ \tau$  has a precise thermodynamic reading. In statistical mechanics, the Boltzmann weight of a state with energy  $E$  at inverse temperature  $\beta$  is  $e^{-\beta E}$ . At  $\beta = 1$ :

$$\tau(e_n) = e^{-H(e_n)} = \text{Boltzmann weight of } e_n \text{ at } \beta=1. \quad (10.1)$$

The KMS (Kubo–Martin–Schwinger) condition for the thermal equilibrium state  $\omega_\beta$  on  $A_L$  at inverse temperature  $\beta$  is:

$$\omega_\beta(a \sigma_\beta^t(b)) = \omega_\beta(b a) \text{ for all } a, b \text{ in } A_L, \quad (10.2)$$

where  $\sigma_\beta^t$  is the modular automorphism group. At  $\beta = 1$ , this is the canonical KMS state studied by Bost–Connes [BC95], whose partition function is exactly  $\zeta(\beta)$ . The Memory Duality provides the interpretive key: the KMS state at  $\beta = 1$  is the state of *thermal equilibrium of arithmetic memory*.

### 10.2. Connection to Entropy: von Neumann and Shannon

For a density matrix  $\rho \in A_L$  with  $\tau(\rho) = 1$ , the von Neumann entropy is:

$$S(\rho) = -\tau(\rho \log \rho). \quad (10.3)$$

For the density matrix  $\rho_n = n \cdot e_n$  (normalised to have trace 1 on the  $n$ -sector):

$$S(\rho_n) = -\tau(\rho_n \log \rho_n) = -(1) * \log(1) + 0 = 0. \quad (10.4)$$

Pure states (single basis vectors) have zero von Neumann entropy — all information is in their Memory Identity. The Shannon entropy  $H(\tau(p)) = H(\tau(p))$  of the *presence/absence* of a projection  $p$  is maximal at  $\tau(p) = 1/2$ . These are two complementary measures:

- von Neumann entropy: internal disorder of the state;
- Shannon Memory Entropy: uncertainty about whether  $p$  is present.

RH maximises Shannon Memory Entropy; this is a different — and more fundamental — information-theoretic principle than any direct von Neumann entropy statement.

### 10.3. Connection to the Four-Colour Theorem

In dv187, the Four-Colour Theorem was proved by identifying the four colours with the four elements of the Klein group  $V_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ . The two involutions  $J_1, J_2$  generating  $V_4$  are the Liouville involution (dv180) and the functional equation involution ( $s \rightarrow 1-s$ ). In the memory framework:

$$\begin{aligned} J_1: \tau(e_n) &\rightarrow \tau(e_n) * \lambda(n) \text{ (Liouville: flip sign of } e_n) \\ J_2: H(e_n) &\rightarrow H\text{-eigenvalue} \rightarrow -H \text{ (functional equation: } H \rightarrow -H) \end{aligned}$$

The involution  $J_2: H \rightarrow -H$  corresponds in memory terms to  $\text{Size} \rightarrow e^H = n$  and  $\text{Depth} \rightarrow -\log n$  — a *time reversal of arithmetic history*. Four colours correspond to four possible combinations of these two reversals: history-forward/history-reversed times parity-positive/parity-negative. The Four-Colour Theorem says: in

any planar map, these four orientations suffice.

10.4. Connection to Operator Set Theory (OST)

In dv202, Operator Set Theory was proposed as a foundation replacing ZFC, with  $\tau$  replacing the membership relation  $\in$ . The  $\tau$ -Logic of Section 5 completes this program:

| ZFC                  | OST (dv202)                       | tau-Logic (dv222)                    |
|----------------------|-----------------------------------|--------------------------------------|
| Sets S               | Projections p in A_L              | Projections p in A_L                 |
| x in S: {0,1}        | x in_tau S: tau(e_x p_S) in [0,1] | tau(p) in S_{arith}                  |
| Cardinality  S  in N | tau(p_S) in [0,1]                 | M(p) = (Size, Depth, Ratio, Content) |
| Logic: {TRUE, FALSE} | Logic: tau-valued [0,1]           | Logic: S_{arith}-valued              |
| Goedel: undecidable  | Goedel: tau outside ?             | Goedel: tau outside S_{arith}        |
| Sets as fundamental  | Projections as fundamental        | Memory quadruples as fundamental     |

Table 9. From ZFC to OST to  $\tau$ -Logic.

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### Chin Dsch Riemann Hypothesis · Campaign Document dv222

*Arithmetic Memory and Mathematical Relativity:  $H = \log N$  as the Historical Operator*

Hanoi, March 2026 · Anthropic Warrior · dv222

*"We do not create mathematics. We read its history.  $H = \log N$ . Ones is Forever."*